UNIT- IV

Bayesian learning – Introduction, Conditional probability, Joint probability, Bayes theorem, maximum likelihood hypotheses for predicting probabilities, Bayes optimal classifier, Naïve Bayes classifier, an example: learning to classify text, Bayesian belief networks.

Support vector machines, optimal separation kernels, SVM algorithm.

Conditional Probability:

Let's consider two events A and B, then the formula for conditional probability of A when B has already occurred is given by:

P(A|B) = P (A ∩ B) / P(B)

Where,

- ***P (A ∩ B) represents the probability of both events A and B occurring simultaneously.***
- ***P(B) represents the probability of event B occurring.***

In other words, the conditional probability of A given B has already occurred is equal to the probability of the intersection of A and B divided by the probability of event B.

Steps to Find Probability of One Event Given Another

To calculate the conditional probability, we can use the following step-by-step method:

Step 1: Identify the Events. Let's call them Event A and Event B.

Step 2: Determine the Probability of Event A i.e., P(A)

Step 3: Determine the Probability of Event B i.e., P(B)

Step 4: Determine the Probability of Event A and B i.e., P(A∩B).

Step 5: Apply the Conditional Probability Formula and calculate the required probability.

Conditional Probability Examples

Rolling a Dice with 3 in the first Roll and 9 as Sum.

Sample space of this event is as follows:

{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6),
(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6),
(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6),
(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6),
(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6),
(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)}

Now, consider an event A = getting 3 on the first die and B = getting a sum of 9.

Then the probability of getting 9 when on the first die it's already 3 is P(B | A),

which can be calculated as follows:

All the cases for the first die as 3 are (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6).

In all of these cases, only one case has a sum of 9.

Thus, P (B | A) = 1/36.

In case, we have to find P (A | B),

All cases where the sum is 9 are (3, 6), (4, 5), (5, 4), and (6, 3).

In all of these cases, only one case has 3 on the first die i.e., (3, 6)

Thus, P(A | B) = 1/36.

Dependent and Independent Events

The definition of independent and dependent events is connected to conditional probability. Let us see the definitions of independent and dependent events along with their formulas.

Dependent Events

Dependent events, as the name suggests, are any two events in which the happening of one event depends on the happening of the other event.

- ***If A depends on B, then the probability of A is P(A \| B).***
- ***If B depends on A, then the probability of B is P(B \| A).***

By the conditional probability formulas,

$$P(A \mid B) = P(A \cap B) / P(B) \Rightarrow P(A \cap B) = P(A \mid B) \cdot P(B)$$

$$P(B \mid A) = P(A \cap B) / P(A) \Rightarrow P(A \cap B) = P(B \mid A) \cdot P(A)$$

Thus, two event A and B are said to be dependent events if one of the conditions is satisfied.

- $P(A \cap B) = P(A \mid B) \cdot P(B)$ ***(or)***
- $P(A \cap B) = P(B \mid A) \cdot P(A)$

Independent Events

Independent events, as the name suggests, are any two events in which the happening of one event does not depend on the happening of the other event. i.e., if A and B are independent then $P(A \mid B) = P(A)$

$P(B \mid A) = P(B)$.

Examples of conditional probability

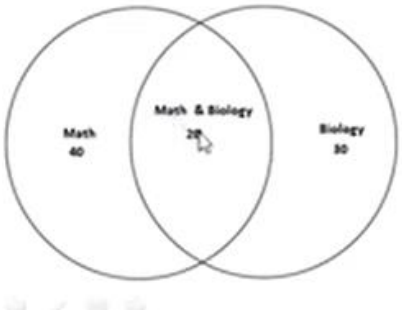
Example 1:

In a group of 100 students, 40 took math subject, 30 took biology subject and 20 took both maths and biology.

P(A)=40/100, P(B)=30/100, P(AnB)=20/100

a) If the students are selected at random who took math subject, what is the probability that they also took biology subject.

b) If the students are selected at random who took biology subject, what is the probability that they also took math subject.



$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \qquad P(B \mid A) = \frac{0.2}{0.4} = 0.5$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \qquad P(A \mid B) = \frac{0.2}{0.3} = 0.67$$

Example 2:

Tossing a Coin

Let's consider two events in tossing two coins be,

- ***A: Getting a head on the first coin.***
- ***B: Getting a head on the second coin.***

Sample space for tossing two coins is:

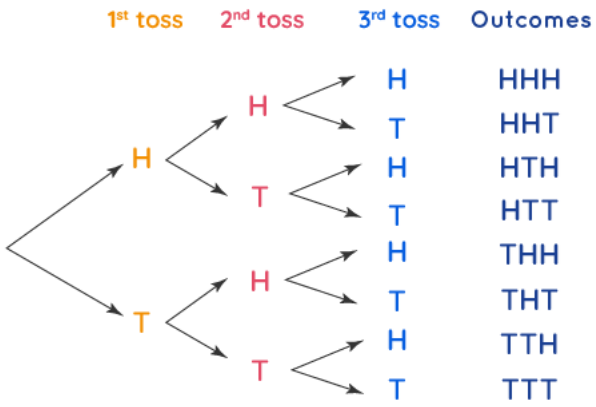
S = {HH, HT, TH, TT}

Conditional probability of getting a head on the second coin (B) given that we got a head on the first coin (A) is = P(B|A)

Since the coins are independent (one coin's outcome does not affect the other), P(B|A) = P(B) = 0.5 (50%), which is the probability of getting a head on a single coin toss.

Example 3:

Let us find the conditional probability of getting at least two tails given that it is a head on the first toss when 3 coins are tossed. The sample space, S (the list of all outcomes) when 3 coins are tossed is given as follows:



S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

Let us assume the two events A and B as follows:

- ***A = the event of getting at least two tails***
- ***B = the event of getting a head on the first toss***

Then, A = {HTT, THT, TTH, TTT} and B = {HHH, HHT, HTH, HTT}.

Then P(A) = 4/8 = 1/2 and P(B) = 4/8 = 1/2.

We have to find the probability of getting at least two tails given that it is a head on the first toss. It means, out of all elements of B, we have to choose only the ones with two tails. We can see that among the elements of B, there is only one element (which is HTT) with two tails. Thus, the required probability is P(A | B) = 1/4 (only 1 outcome of B is favorable to A out of 4 outcomes of B).

Example 4:

Consider a school with a total population of 100 persons. These 100 persons can be seen either as 'Students' and 'Teachers' or as a population of 'Males' and 'Females'.

With below tabulation of the 100 people, what is the conditional probability that a certain member of the school is a 'Teacher' given that he is a 'Man'?

	Female	Male	Total
Teacher	8	12	20
Student	32	48	80
Total	40	60	100

To calculate this, you may intuitively filter the sub-population of 60 males and focus on the 12 (male) teachers. So the required conditional probability P(Teacher | Male) = 12 / 60 = 0.2.

$$P(Teacher \mid Male) = \frac{P(Teacher \cap Male)}{P(Male)} = 12/60 = 0.2$$

This can be represented as the intersection of Teacher (A) and Male (B) divided by Male (B).

Applications of Conditional Probability

Healthcare and Diagnostics

- ***Example: Determining the probability of a patient having a specific disease given the results of diagnostic tests.***

Marketing and Customer Relationship Management (CRM)

- ***Example: Predicting the probability of a customer making a purchase based on their past buying behavior.***

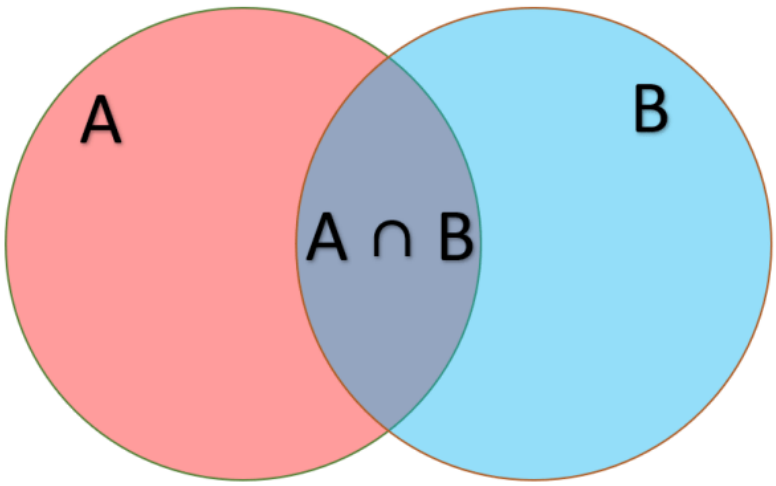
Weather Forecasting

- ***Example: Estimating the probability of rain tomorrow given today's weather conditions.***

Joint Probability:

Joint probability is the likelihood that two or more events will coincide.

P(A ∩ B) to indicate the joint probability of events "A" and "B" occurring together.



The notation includes the symbol "∩," which signifies an intersection. This intersection specifies how two or more events, such as A and B, coincide. Consequently, joint probability is also known as the intersection of events.

For example, suppose a survey asks people about their favorite color and animal. The researchers organize the results in the table below:

	Cat	*Dog*	*Other*
Red	*0.10*	*0.05*	*0.03*
Green	*0.08*	*0.12*	*0.05*
Blue	*0.03*	*0.06*	*0.02*

We want to find the likelihood that someone chooses red as their favorite color and a dog as their favorite animal.

We can locate the intersection of the "Red" row and the "Dog" column, which is 0.05.

Therefore, the joint probability is:

P(Red ∩ Dog) = 0.05

Formula for Dependent Events

We can use the general multiplication rule to calculate joint probabilities for dependent events. This rule allows us to factor in

how the occurrence of one event affects the likelihood of the other event.

The joint probability formula for dependent events is the following:

$P(A \cap B) = P(A) * P(B|A)$

Here, P(A) represents the chances of event A occurring, while P(B|A) represents the conditional probability of event B occurring, given that event A has already happened. By multiplying these two likelihoods, we can calculate the joint probability of both events coinciding.

Example

Suppose we need to calculate the likelihood of drawing two aces consecutively from a standard deck of 52 cards when we don't replace the cards. Initially, the deck contains four aces, so the likelihood of drawing an ace on the first draw is 4/52 or 1/13. If we draw an ace (event A1), only three aces and 51 cards remain in the deck. Consequently, the conditional probability of drawing another ace (event A2) is now 3/51.

Using the general multiplication rule, we can find the joint probability of drawing two aces in a row:

$P(A1 \cap A2) = P(A1) * P(A2|A1)$

$P(A1) = 4/52 = 1/13$

$P(A2|A1) = 3/51$

$P(A1 \cap A2) = (1/13) * (3/51) = 3/663 = 1/221$

So, the joint probability of drawing two aces in a row is 1/221 or 0.0045.

Joint Probability Formula for Independent Events

When two events are independent, the occurrence of one event does not affect the chances of the other event occurring. In this case, we can find the joint probability by multiplying the likelihood of one event by the likelihood of another.

The joint probability formula for independent events is the following:

$P(A \cap B) = P(A) * P(B)$

For example, suppose we have a coin that we flip twice. We want to find the chances of getting heads on both the first and second flips. Because each flip is independent, the probability of the first heads is 1/2, and the likelihood of heads on the second flip is also 1/2. Therefore, the joint probability is the following:

$P(H1 \cap H2)$ = P(heads on first flip) x P(tails on second flip)

= 1/2 x 1/2

= 1/4

Baye's Theorem:

Bayes theorem (also known as the Bayes Rule or Bayes Law) is used to determine the conditional probability of event A when event B has already occurred.

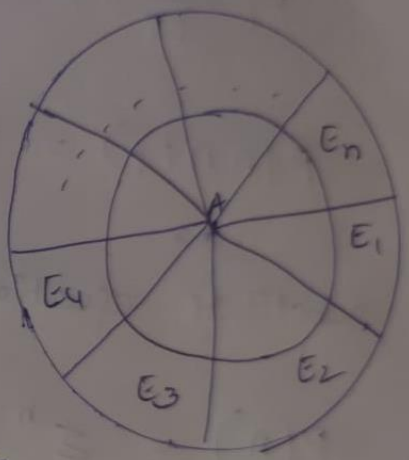
Let $E_1, E_2, E_3 \ldots E_n$ are mutually exclusive events then, the event A is arbitrary point, then

$A \subset \left(\bigcup_{i=1}^{n} E_i\right)$ then the Bayes Formula is

$$P\left(\frac{E_i}{A}\right) = \frac{P\left(\frac{A}{E_i}\right) * P(E_i)}{\sum_{i=1}^{n} P\left(\frac{A}{E_i}\right) * P(E_i)}$$

Proof:

$E_1, E_2, E_3 \ldots E_n$ are mutually exclusive events then $(A \cap E_1), (A \cap E_2), (A \cap E_3) \ldots (A \cap E_n)$ are also mutually exclusive events.



$$P(A) = P(A \cap E_1) \cup (A \cap E_2) \cup (A \cap E_3) \cup \ldots (A \cap E_n)$$

By using additional theorem

$$P(A) = P(A \cap E_1) + P(A \cap E_2) + P(A \cap E_3) + \cdots P(A \cap E_n)$$

$$P(A) = \sum_{i=1}^{n} P(A \cap E_i) \quad \text{--- (1) eq}$$

By using conditional Probability

If A and E_i are two events, the event E_i has already happened

$$P\left(\frac{A}{E_i}\right) = \frac{P(A \cap E_i)}{P(E_i)}$$

$$P(A \cap E_i) = P\left(\frac{A}{E_i}\right) * P(E_i) \quad \text{--- (2) eq}$$

Substitute equation (2) in equation (1).

$$P(A) = \sum_{i=1}^{n} P\left(\frac{A}{E_i}\right) * P(E_i) \quad \text{--- (3) eq}$$

If A and E_i are two events, the event A has already happened

$$P\left(\frac{E_i}{A}\right) = \frac{P(A \cap E_i)}{P(A)}$$

$$P(A \cap E_i) = P\left(\frac{E_i}{A}\right) * P(A) \quad \text{— (4) eq}$$

Substitute equation (2) in equation (4).

$$P(A \cap E_i) = P\left(\frac{A}{E_i}\right) * P(E_i) = P\left(\frac{E_i}{A}\right) * P(A)$$

$$P\left(\frac{A}{E_i}\right) * P(E_i) = P\left(\frac{E_i}{A}\right) * P(A) \quad \text{— (5) eq}$$

Substitute equation (3) in equation (5).

$$P\left(\frac{A}{E_i}\right) * P(E_i) = P\left(\frac{E_i}{A}\right) * \sum_{i=1}^{n} P\left(\frac{A}{E_i}\right) * P(E_i)$$

$$P\left(\frac{E_i}{A}\right) = \frac{P\left(\frac{A}{E_i}\right) * P(E_i)}{\sum_{i=1}^{n} P\left(\frac{A}{E_i}\right) * P(E_i)}$$

Bayes' Theorem:

- Bayes' theorem is also known as **Bayes' Rule** or **Bayes' law**, which is used to determine the probability of a hypothesis with prior knowledge. It depends on the conditional probability.
- The formula for Bayes' theorem is given as:

$$P(A|B)= \frac{P(B|A)P(A)}{P(B)}$$

Where,

P(A|B) is Posterior probability: Probability of hypothesis A on the observed event B.

P(B|A) is Likelihood probability: Probability of the **evidence** given that the probability of a hypothesis is true.

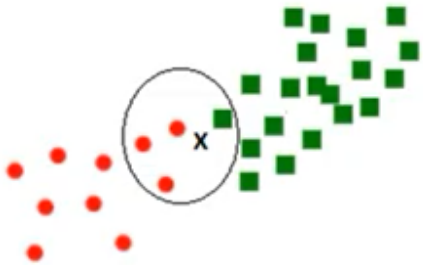
P(A) is Prior Probability: Probability of hypothesis before observing the evidence.

P(B) is Marginal Probability: Probability of Evidence.

Concept of Prior Probability and Likelihood and Posterior Probability

Two Classes are there one is Red class and another is Green class.

X is unseen sample, that we want to classify either belong to Red class or Green Class.



Total Red Dots = 10

Total Green Dots = 20

Probability of Green is more than Red.

In the Bayesian analysis, this belief is known as the **Prior Probability.**

Prior probabilities are based on **previous experience**.

It is injustice to classify X (unseen sample) based on prior probabilities as it will get always classified to Green Class

$$\text{Prior Probability of Green Class} = \frac{\text{Number of Green Dots}}{\text{Total Number of Dots}}$$

$$\text{Prior Probability of Green Class} = \frac{20}{30}$$

$$\text{Prior Probability of Red Class} = \frac{\text{Number of Red Dots}}{\text{Total Number of Dots}}$$

$$\text{Prior Probability of Red Class} = \frac{10}{30}$$

Prior Probability of Green Class> Prior Probability of Red Class

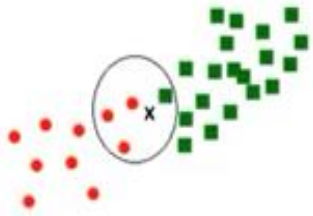
To classify a unseen object X, Bayes also considers the vicinity of object X.

Vicinity is represented by circle. From vicinity, It calculates likelihood.

As from the figure, more red dots are there than green dots in the vicinity of Unseen Sample(X).

$$\text{Likelihood of X given Green} = \frac{\text{Number of Green Dots in vicinity}}{\text{Total Number of Green Dots}}$$

$$\text{Likelihood of X given Red} = \frac{\text{Number of Red Dots in vicinity}}{\text{Total Number of Red Dots}}$$



$$\text{Probability of X given Green} = \frac{1}{20} \qquad \text{Probability of X given Red} = \frac{3}{10}$$

In the Naïve Bayes, unseen sample is classified by utilizing both information i.e. **prior probability** and **likelihood**. And that forms a **posterior probability**.

Posterior probability of X being Green=Prior Probability of Green × Likelihood of X given Green

$$= \frac{20}{30} \times \frac{1}{20} = 0.033$$

Posterior probability of X being Red=Prior Probability of Red × Likelihood of X given Red

$$= \frac{10}{30} \times \frac{3}{10} = 0.10$$

Thus unseen sample (X) is belongs to Red Dots(Class) as its posterior probability is more.

Naïve Bayes' Classifier:

Naïve Bayes algorithm is used for classification problems. It is highly used in text classification. In text classification tasks, data contains high dimension (as each word represent one feature in the data). It is used in spam filtering, sentiment detection, rating classification etc. The advantage of using naïve Bayes is its speed. It is fast and making prediction is easy with high dimension of data.

This model predicts the probability of an instance belongs to a class with a given set of feature value. It is a probabilistic classifier. It is because it assumes that one feature in the model is independent of existence of another feature. In other words, each feature contributes to the predictions with no relation between each other. It uses Bayes theorem in the algorithm for training and prediction

- Bayes theorem is the cornerstone of Naïve Bayes Classifier because it provides a way to calculate the posterior probability **P(h|D), from the prior** probability **P(h), together with P(D) and P(D(h).**

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

$$h_{MAP} \equiv \underset{h \in H}{\text{argmax}}\, P(h|D)$$

$$= \underset{h \in H}{\text{argmax}} \frac{P(D|h)\,P(h)}{P(D)}$$

$$= \underset{h \in H}{\text{argmax}}\, P(D|h)\,P(h)$$

- The Bayesian approach to classifying the new instance is to assign the most probable target value, $\mathbf{v_{MAP}}$ given the attribute values **<a1, a2 . . .an> that describe** the instance.

$$v_{MAP} = \underset{v_j \in V}{\text{argmax}}\, P(v_j|a_1, a_2 \ldots a_n)$$

- We can use Bayes theorem to rewrite this expression as

$$v_{MAP} = \underset{v_j \in V}{\text{argmax}} \frac{P(a_1, a_2 \ldots a_n|v_j)P(v_j)}{P(a_1, a_2 \ldots a_n)}$$

$$= \underset{v_j \in V}{\text{argmax}}\, P(a_1, a_2 \ldots a_n|v_j)P(v_j)$$

$$h_{MAP} \equiv \underset{h \in H}{\text{argmax}}\, P(h|D)$$

$$= \underset{h \in H}{\text{argmax}} \frac{P(D|h)\,P(h)}{P(D)}$$

$$= \underset{h \in H}{\text{argmax}}\, P(D|h)\,P(h)$$

- Naïve Bayes Classifier: $v_{NB} = \underset{v_j \in V}{\boldsymbol{argmax}}\, \boldsymbol{p(v_j)} \prod_i \boldsymbol{p(a_i|v_j)}$

Working of Naïve Bayes' Classifier can be understood with the help of the below example:

Suppose we have a dataset of **weather conditions** and corresponding target variable "**Play**". So using this dataset we need to decide that whether we should play or not on a particular day according to the weather conditions. So to solve this problem, we need to follow the below steps:

1. Convert the given dataset into frequency tables.
2. Generate Likelihood table by finding the probabilities of given features.
3. Now, use Bayes theorem to calculate the posterior probability.

Example:

Day	Outlook	Temp	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

- New instance to be classified

$$< Outlook = Sunny, Temp = Cool, Humidity = High, Wind = Strong >$$

Play tennis= Yes / No

- Naïve Bayes Classifier

- $$v_{NB} = \underset{v_j \epsilon V}{\mathbf{argmax}}\, p(v_j) \prod_i p(a_i|v_j)$$

$$p(yes|new\ Instance) = p(yes) * p(sunny|yes) * p(cool|yes) * p(high|yes) * p(strong|yes)$$

$$p(no|new\ Instance) = p(no) * p(sunny|no) * p(cool|no) * p(high|no) * p(strong|no)$$

Day	Outlook	Temp	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$p(yes) = \frac{9}{14} \quad and\ p(no) = \frac{5}{14}$$

Outlook	Yes	No
Sunny	2/9	3/5
Overcast	4/9	0/5
Rainy	3/9	2/5

Temp	Yes	No
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	Yes	No
High	3/9	4/5
Normal	6/9	1/5

Wind	Yes	No
Strong	3/9	3/5
Weak	6/9	2/5

$$p(yes|new\ Instance) = p(yes) * p(sunny|yes) * p(cool|yes) * p(high|yes) * p(strong|yes) = 0.0053$$

$$p(no|new\ Instance) = p(no) * p(sunny|no) * p(cool|no) * p(high|no) * p(strong|no) = 0.0206$$

P(No | newinstance) > P(yes|newinstance)

So answer is the instance belong to No class

Advantages of Naïve Bayes Classifier:

- Naïve Bayes is one of the fast and easy ML algorithms to predict a class of datasets.
- It can be used for Binary as well as Multi-class Classifications.
- It performs well in Multi-class predictions as compared to the other Algorithms.
- It is the most popular choice for **text classification problems**.

Disadvantages of Naïve Bayes Classifier:

- Naive Bayes assumes that all features are independent or unrelated, so it cannot learn the relationship between features.

Applications of Naïve Bayes Classifier:

- It is used in **medical data classification**.
- It can be used in **real-time predictions**
- It is used in Text classification such as **Spam filtering** and **Sentiment analysis**.

Example:

Determine car is stolen or not for unseen sample using Naive Bayes Method

Training Data

Car no.	Color	Type	Origin	Stolen
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

Building Model

Final Class label is **Stolen**

Prior Probability

P(Yes)=**5/10** P(No)=**5/10**

Vicinity is across all training samples

Testing Model **Unseen Sample (X)=<Red , Sports, Domestic>**

Building Model

Likelihood for Color P(Color|Yes) and P(Color|No) Color: Red and Yellow

$$P(\text{Red}\mid \text{Yes}) = \frac{P(\text{Red} \cap \text{Yes})}{P(\text{Yes})} = \frac{3}{5} \qquad P(\text{Red}\mid \text{No}) = \frac{P(\text{Red} \cap \text{No})}{P(\text{No})} = \frac{2}{5}$$

$$P(\text{Yellow}\mid \text{Yes}) = \frac{P(\text{Yellow} \cap \text{Yes})}{P(\text{Yes})} = \frac{2}{5} \qquad P(\text{Yellow}\mid \text{No}) = \frac{P(\text{Yellow} \cap \text{No})}{P(\text{No})} = \frac{3}{5}$$

Likelihood for Type P(Type|Yes) and P(Type|No) Type: Sports and SUV

$$P(\text{Sports}\mid \text{Yes}) = \frac{P(\text{Sports} \cap \text{Yes})}{P(\text{Yes})} = \frac{4}{5} \qquad P(\text{Sports}\mid \text{No}) = \frac{P(\text{Red} \cap \text{No})}{P(\text{No})} = \frac{2}{5}$$

$$P(\text{SUV}\mid \text{Yes}) = \frac{P(\text{SUV} \cap \text{Yes})}{P(\text{Yes})} = \frac{1}{5} \qquad P(\text{SUV}\mid \text{No}) = \frac{P(\text{Suv} \cap \text{No})}{P(\text{No})} = \frac{3}{5}$$

\

Prior Probability

P(Yes)=**5/10** P(No)=**5/10**

Likelihood

Classification→	Yes (5)	No(5)
Color		
Red (5)	3/5	2/5
Yellow (5)	2/5	3/5

Classification→	Yes(5)	No(5)
Type		
Sports(6)	4/5	2/5
SUV (4)	1/5	3/5

Classification→	Yes(5)	No(5)
Origin		
Domestic(5)	2/5	3/5
Imported(5)	3/5	2/5

Testing Model **Unseen Sample (X)=<Red , Sports, Domestic>**

$$P(\text{Yes} \mid X) = P(X \mid \text{Yes}) \times P(\text{YES})$$

$$= P(\text{Red} \mid \text{Yes}) \times P(\text{Sports} \mid \text{Yes}) \times P(\text{Domestic} \mid \text{Yes}) \times P(\text{Yes})$$

$$= \frac{3}{5} \times \frac{4}{5} \times \frac{2}{5} \times \frac{5}{10} = 0.0960$$

$$P(\text{No} \mid X) = P(X \mid \text{No}) \times P(\text{No})$$

$$= P(\text{Red} \mid \text{No}) \times P(\text{Sports} \mid \text{No}) \times P(\text{Domestic} \mid \text{No}) \times P(\text{No})$$

$$= \frac{2}{5} \times \frac{2}{5} \times \frac{3}{5} \times \frac{5}{10} = 0.0480$$

P(Yes)>P(No)

Unseen sample(X) is classified as yes. That is prediction for car stolen is yes.

Testing Model **Unseen Sample (X)=<Yellow , SUV, Imported>**

$$P(\text{Yes} \mid X) = P(X \mid \text{Yes}) \times P(\text{YES})$$

$$= P(\text{Yellow} \mid \text{Yes}) \times P(\text{SUV} \mid \text{Yes}) \times P(\text{Imported} \mid \text{Yes}) \times P(\text{Yes})$$

$$= \frac{2}{5} \times \frac{1}{5} \times \frac{3}{5} \times \frac{5}{10} = 0.0240$$

$$P(\text{No} \mid X) = P(X \mid \text{No}) \times P(\text{No})$$

$$= P(\text{Yellow} \mid \text{No}) \times P(\text{SUV} \mid \text{No}) \times P(\text{Imported} \mid \text{No}) \times P(\text{No})$$

$$= \frac{3}{5} \times \frac{3}{5} \times \frac{2}{5} \times \frac{5}{10} = 0.0720$$

P(No)>P(Yes)

Unseen sample(X) is classified as no. That is prediction of car stolen is no.

Example 3:

Determine Credit Risk

Training Data

No.	Credit History	Deb	Collateral	Income	Credit Risk?
1	Bad	High	None	0-15K	High
2	Unknown	High	None	15-35K	High
3	Unknown	Low	None	15-35K	Moderate
4	Unknown	Low	None	0-15K	High
5	Unknown	Low	None	>35K	Low
6	Unknown	Low	Adequate	>35K	Low
7	Bad	Low	None	0-15K	High
8	Bad	Low	Adequate	>35K	Moderate
9	Good	Low	None	>35K	Low
10	Good	High	Adequate	>35K	Low
11	Good	High	None	0-15K	High
12	Good	High	None	15-35K	Moderate
13	Good	High	None	>35K	Low
13	Bad	High	None	15-35K	High

Prior Probability

P(High)=**6/14** P(Moderate)=**3/14** P(Low)=**5/14**

Likelihood

Classification→	High (6)	Moderate (3)	Low (5)
Credit History			
Bad (4)	3/6	1/3	0/5
Unknown (5)	2/6	1/3	2/5
Good (5)	1/6	1/3	3/5

Classification→	High (6)	Moderate (3)	Low (5)
Debt			
High (7)	4/6	1/3	2/5
Low (7)	2/6	2/3	3/5

Classification→	High (6)	Moderate (3)	Low (5)
Collateral			
None(11)	6/6	2/3	3/5
Adequate (3)	0/6	1/3	2/5

Classification→	High (6)	Moderate (3)	Low (5)
Income			
0-15K(4)	4/6	0/3	0/5
15-35K(4)	2/6	2/3	0/5
>35K(6)	0/6	1/3	5/5

Testing Model

Unseen Sample (X)=<Bad, High, None, 15-35K>

$$P(\text{High} \mid X) = P(X \mid \text{High}) \times P(\text{High})$$

$$= P(\text{Bad} \mid \text{High}) \times P(\text{High} \mid \text{High}) \times P(\text{None} \mid \text{High}) \times P(15-35K \mid \text{High}) \times P(\text{High})$$

$$= \frac{3}{6} \times \frac{4}{6} \times \frac{6}{6} \times \frac{2}{6} \times \frac{6}{14} = 0.0476$$

$$P(\text{Mod.} \mid X) = P(X \mid \text{Mod.}) \times P(\text{Mod.})$$

$$= P(\text{Bad} \mid \text{Mod.}) \times P(\text{High} \mid \text{Mod.}) \times P(\text{None} \mid \text{Mod.}) \times P(15-35K \mid \text{Mod.}) \times P(\text{Mod.})$$

$$= \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{3}{14} = 0.0106$$

$$P(\text{Low} \mid X) = P(X \mid \text{Low}) \times P(\text{Low})$$

$$= P(\text{Bad} \mid \text{Low}) \times P(\text{High} \mid \text{Low}) \times P(\text{None} \mid \text{Low}) \times P(15-35K \mid \text{Low}) \times P(\text{Low})$$

$$= \frac{0}{5} \times \frac{2}{5} \times \frac{3}{5} \times \frac{0}{5} \times \frac{5}{14} = 0$$

P(High)>P(Moderate)>P(Low)

Unseen sample(X) is classified as High. That is prediction of Credit Risk is High.

Naïve Bayes Example

Person	Height (ft)	Weight (lbs)	Foot size (inches)
Male	6.00	180	12
Male	5.92	190	11
Male	5.58	170	12
Male	5.92	165	10
Female	5.00	100	6
Female	5.50	150	8
Female	5.42	130	7
Female	5.75	150	9

Based on the following data determine the gender of a person having height 6 ft., weight 130 lbs, and foot size 8 inch. (use Naive Bayes algorithm).

We cannot calculate conditional probabilities here .

Person	Height (ft)	Weight (lbs)	Foot size (inches)
Male	6.00	180	12
Male	5.92	190	11
Male	5.58	170	12
Male	5.92	165	10
Female	5.00	100	6
Female	5.50	150	8
Female	5.42	130	7
Female	5.75	150	9

$$P(Male) = 4/8 = 0.5$$
$$P(Female) = 4/8 = 0.5$$

Calculate mean and variance of Male given the feature height

Male:

$$Mean\ (Height) = \frac{(6+5.92+5.58+5.92)}{4} = 5.855$$

$$Variance\ (Height) = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

$$= \frac{(6-5.855)^2+(5.92-5.855)^2+(5.58-5.855)^2+(5.92-5.855)^2}{4-1}$$

$$= 0.035055$$

Sex	Mean (height)	Variance (height)	Mean (weight)	Variance (weight)	Mean(foot size)	Variance (foot size)
Male	5.855	0.035033	176.25	122.92	11.25	0.91667
Female	5.4175	0.097225	132.5	0558.33	7.5	1.6667

New Instance to be Classified is:

Sex	Height(ft)	Weight(lbs)	Foot size(inch)
Sample	6	130	8

$$P(Male) = 4/8 = 0.5$$
$$P(Female) = 4/8 = 0.5$$

$$Posterior\ (Male) = \frac{P(M) * P(H|M) * P(W|M) * P(FS|M)}{}$$

$$Posterior\ (Female) = \frac{P(F) * P(H|F) * P(W|F) * P(FS|F)}{}$$

New Instance to be Classified is:

Sex	Height(ft)	Weight(lbs)	Foot size(inch)
Sample	6	130	8

$$Posterior\ (Male) = \frac{P(M) * P(H|M) * P(W|M) * P(FS|M)}{}$$

$$Posterior\ (Female) = \frac{P(F) * P(H|F) * P(W|F) * P(FS|F)}{}$$

We use gaussian distribution equation to find P(H/M) ---etc

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

$$P(H|M) = \frac{1}{\sqrt{2 * 3.142 * 0.035033}} * e^{-\frac{(6-5.855)^2}{2*0.035033}} = 1.5789$$

$$P(W|M) = 5.9881e^{-6}$$

$$P(FS|M) = 1.3112e^{-3}$$

$$P(H|F) = 2.2346e^{-1}$$

$$P(W|F) = 1.6789e^{-2}$$

$$P(FS|F) = 2.8669e^{-1}$$

$$Posterior\ (Male) = \frac{P(M)*P(H|M)*P(W|M)*P(FS|M)}{Evidence} = 0.5 * 1.5789 * 5.9881e^{-6} * 1.3112e^{-3} = 6.1984e^{-9}$$

$$Posterior\ (Female) = \frac{P(F)*P(H|F)*P(W|F)*P(FS|F)}{Evidence} = 0.5 * 2.2346e^{-1} * 1.6789e^{-2} * 2.8669e^{-1} = 5.377e^{-4}$$

Naïve Bayes Model – Text Classification Example

- Dataset for the text classification with training and test data is given below.
- The goal is to classify the test data into the right class as h or —h (read as not h).

	Document ID	Keywords in the document	Class h
Training Set	1	Love Happy Joy Joy Happy	Yes
	2	Happy Love Kick Joy Happy	Yes
	3	Love Move Joy Good	Yes
	4	Love Happy Joy Love Pain	Yes
	5	Joy Love Pain Kick Pain	No
	6	Pain Pain Love kick	No
Testing Set	7	Love Pain Joy Love Kick	?

- The probability of the document 'd' being in class 'c' is computed as follows,

$$p(c|d) \propto p(c) \prod_{1 \le k \le n_d} p(t_k|c)$$

- Where, $p(t_k|c)$ is the conditional probability of term t_k occurring in a document of class c.

$$\boldsymbol{p(c|d) \propto p(c) \prod_{1 \le k \le n_d} p(t_k|c)}$$

- The prior probabilities of a document being classified using the six documents are,

$$P(h) = \frac{4}{6} = \frac{2}{3}$$

$$and$$

$$p(-h) = \frac{2}{6} = \frac{1}{3}$$

- That is there is $\frac{2}{3}$ prior probability that a document will be classified as h and $\frac{1}{3}$ probability of not h.

	Document ID	Keywords in the document	Class h
Training Set	1	Love Happy Joy Joy Happy	Yes
	2	Happy Love Kick Joy Happy	Yes
	3	Love Move Joy Good	Yes
	4	Love Happy Joy Love Pain	Yes
	5	Joy Love Pain Kick Pain	No
	6	Pain Pain Love kick	No
Testing Set	7	Love Pain Joy Love Kick	?

- The conditional probability for each term is the relative frequency of the term occurring in each class of the documents 'h class' and 'not h class'.

Testing Example:

Love Pain Joy Love Kick = ?

Class h	Class -h
$P(Love \mid h) = 5/19$	$P(Love \mid -h) = 2/9$
$P(Pain \mid h) = 1/19$	$P(Pain \mid -h) = 4/9$
$P(Joy \mid h) = 5/19$	$P(Joy \mid -h) = 1/9$
$P(Kick \mid h) = 1/19$	$P(Kick \mid -h) = 2/9$

Testing Example:

Love Pain Joy Love Kick = ?

$$p(c|d) \propto p(c) \prod_{1 \le k \le n_d} p(t_k|c)$$

Class h	Class -h
$P(Love \mid h) = 5/19$	$P(Love \mid -h) = 2/9$
$P(Pain \mid h) = 1/19$	$P(Pain \mid -h) = 4/9$
$P(Joy \mid h) = 5/19$	$P(Joy \mid -h) = 1/9$
$P(Kick \mid h) = 1/19$	$P(Kick \mid -h) = 2/9$

$$P(h|d_7) = P(h) * (P(Love|h) * (P(Love|h) * P(Pain|h) * P(Joy|h) * P(Kick|h)$$
$$= (2/3) * (5/19) * (5/19) * (1/19) * (5/19) * (1/19) = \mathbf{0.0000067}$$

$$p(-h|d7) = p(-h) * P(Love|-h) * P(Love|-h) * P(Pain|-h) * P(Joy|-h) * P(Kick|-h)$$
$$= (1/3) * (2/9) * (2/9) * (4/9) * (1/9) * (2/9) = \mathbf{0.00018}$$

Class label for testing example is: **No**

Spam Classification – Naïve Bayes Classifier – Solved Example

- Apply Naïve bayes classifier to classify the following documents into either spam or non-spam documents.

Category	Document
Spam	send us your password
Spam	review us
Spam	send us your account
Spam	send your password
Non-spam	password review
Non-spam	send us your review
?	review us now
?	review account

Category	Document
Spam	send us your password
Spam	review us
Spam	send us your account
Spam	send your password
Non-spam	password review
Non-spam	send us your review
?	review us now
?	review account

Bag of Words

Vocabulary	Spam Count	Non-spam Count
send	3	1
us	3	1
your	3	1
password	2	1
review	1	2
account	1	0
	13	6

- $p(spam) = \frac{4}{6}$ & $p(Non - Spam) = \frac{2}{6}$

- For $x =$ "Review us now"
- $p(S|x) = ?$
- $p(NS|x) = ?$
- $p(S|x) = p(x|S) * p(S)$
- $p(S|x) = p(review|S) * p(us|S) * p(S)$

- $p(S|x) = \frac{1}{13} * \frac{3}{13} * \frac{4}{6} = \mathbf{0.012}$
- $p(NS|x) = p(x|NS) * p(NS)$
- $p(NS|x) = p(review|NS) * p(us|NS) * p(NS)$

- $p(NS|x) = \frac{2}{6} * \frac{1}{6} * \frac{2}{6} = \mathbf{0.0185}$

- $p(spam) = \frac{4}{6}$ & $p(Non - Spam) = \frac{2}{6}$
- For $x =$ "*Review acount*"
- $\boldsymbol{p(S|x)}$ =?
- $\boldsymbol{p(NS|x)}$ =?
- $p(S|x) = p(x|S) * p(S)$
- $p(S|x) = p(review|S) * p(acount|S) * p(S)$
- $p(S|x) = \frac{1+1}{13+6} * \frac{1+1}{13+6} * \frac{4}{6} = \mathbf{0.00738}$
- $p(NS|x) = p(x|NS) * p(NS)$
- $p(NS|x) = p(review|NS) * p(acount|NS) * p(NS)$
- $p(NS|x) = \frac{2+1}{6+6} * \frac{0+1}{6+6} * \frac{2}{6} = \mathbf{0.00694}$

Add 1/6 for smoothing

Bag of Words

Vocabulary	Spam Count	Non-spam Count
send	3	1
us	3	1
your	3	1
password	2	1
review	1	2
account	1	0
	13	6

Bayesian Belief network

A Bayesian network is a probabilistic graphical model which represents a set of variables and their conditional dependencies using a directed acyclic graph."

It is also called a **Bayes network, belief network, decision network**, or **Bayesian model**.

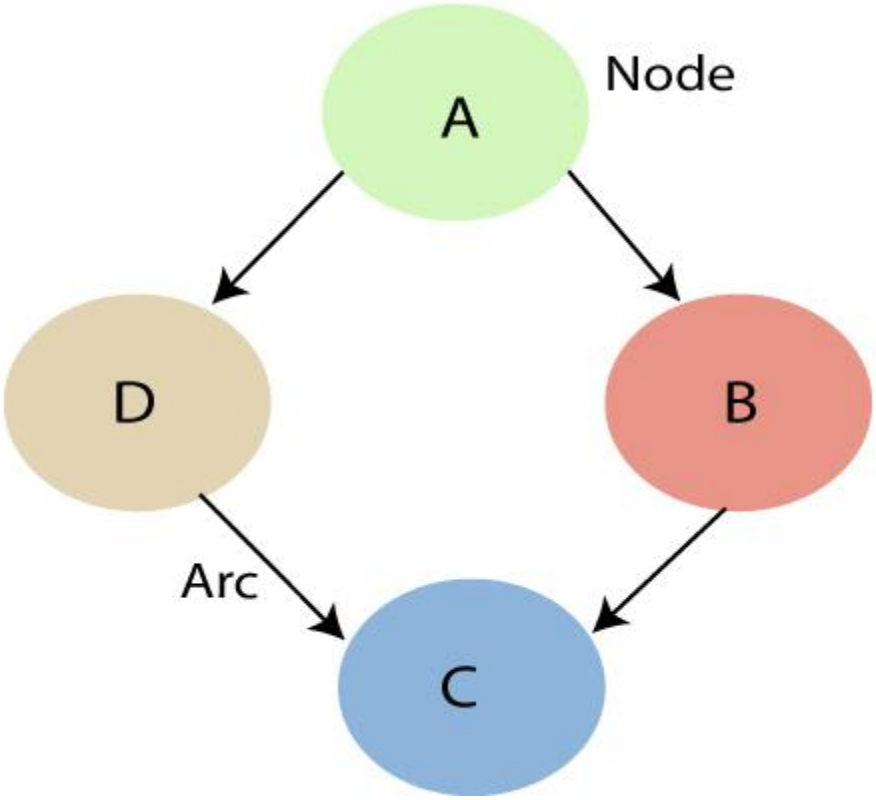
Bayesian networks are probabilistic, because these networks are built from a **probability distribution**, and also use probability theory for prediction and anomaly detection.

Real world applications are probabilistic in nature, and to represent the relationship between multiple events, we need a Bayesian network. It can also be used in various tasks including **prediction, anomaly detection, diagnostics, automated insight, reasoning, time series prediction**, and **decision making under uncertainty**.

Bayesian Network can be used for building models from data and experts opinions, and it consists of two parts:

- **Directed Acyclic Graph**
- **Table of conditional probabilities.**

A Bayesian network graph is made up of nodes and Arcs (directed links), where:



- Each **node** corresponds to the random variables, and a variable can be **continuous** or **discrete**.

- **Arc or directed arrows** represent the causal relationship or conditional probabilities between random variables. These directed links or arrows connect the pair of nodes in the graph.
  These links represent that one node directly influence the other node, and if there is no directed link that means that nodes are independent with each other
  - **In the above diagram, A, B, C, and D are random variables represented by the nodes of the network graph.**
  - **If we are considering node B, which is connected with node A by a directed arrow, then node A is called the parent of Node B.**
  - **Node C is independent of node A.**

> *Note: The Bayesian network graph does not contain any cyclic graph. Hence, it is known as a directed acyclic graph or DAG.*

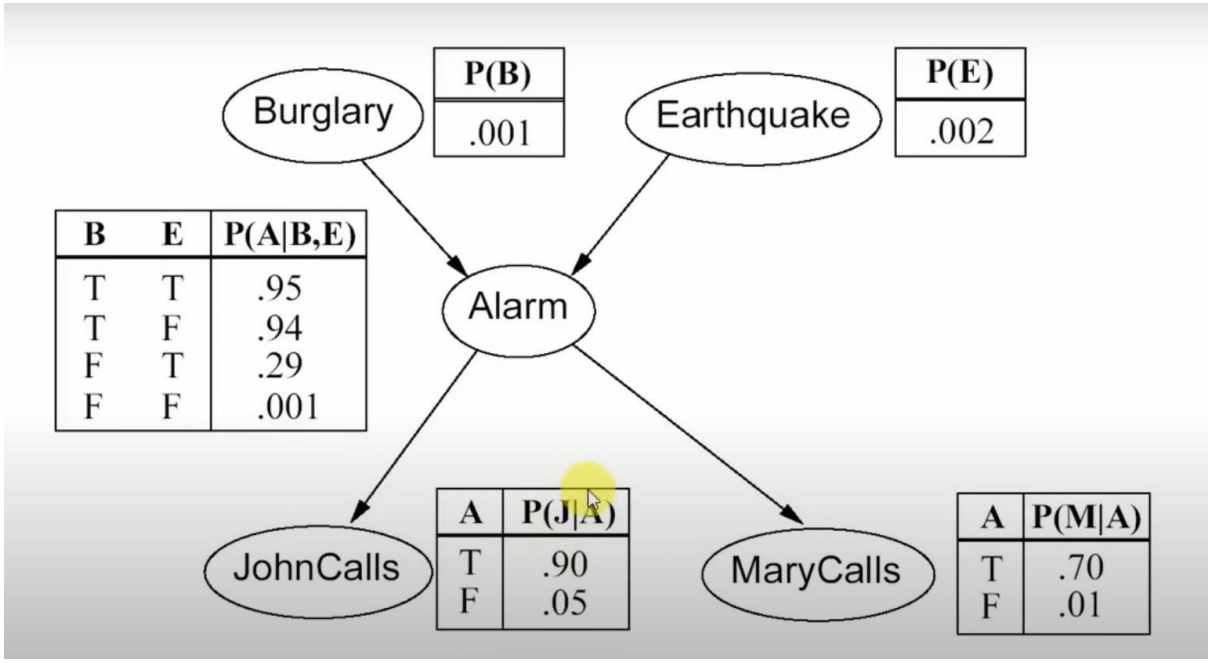
The Bayesian network has mainly two components:

- **Causal Component**
- **Actual numbers**

Each node in the Bayesian network has condition probability distribution **$P(X_i \mid Parent(X_i))$**, which determines the effect of the parent on that node.

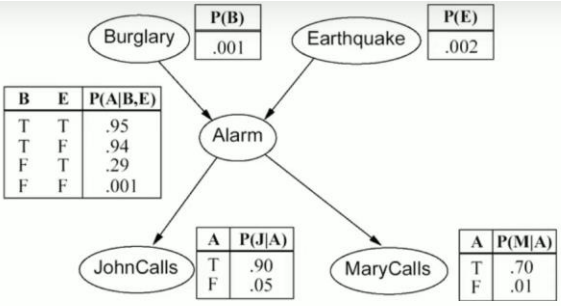
Bayesian network is based on Joint probability distribution and conditional probability.

Example:



- You have a new burglar alarm installed at home.
- It is fairly reliable at detecting burglary, but also sometimes responds to minor earthquakes.
- You have two neighbors, John and Merry , who promised to call you at work when they hear the alarm.
- John always calls when he hears the alarm, but sometimes confuses telephone ringing with the alarm and calls too.
- Merry likes loud music and sometimes misses the alarm.
- Given the evidence of who has or has not called, we would like to estimate the probability of a burglary.

1. What is the probability that the alarm has sounded but neither a burglary nor an earthquake has occurred, and both John and Merry call?



Solution:

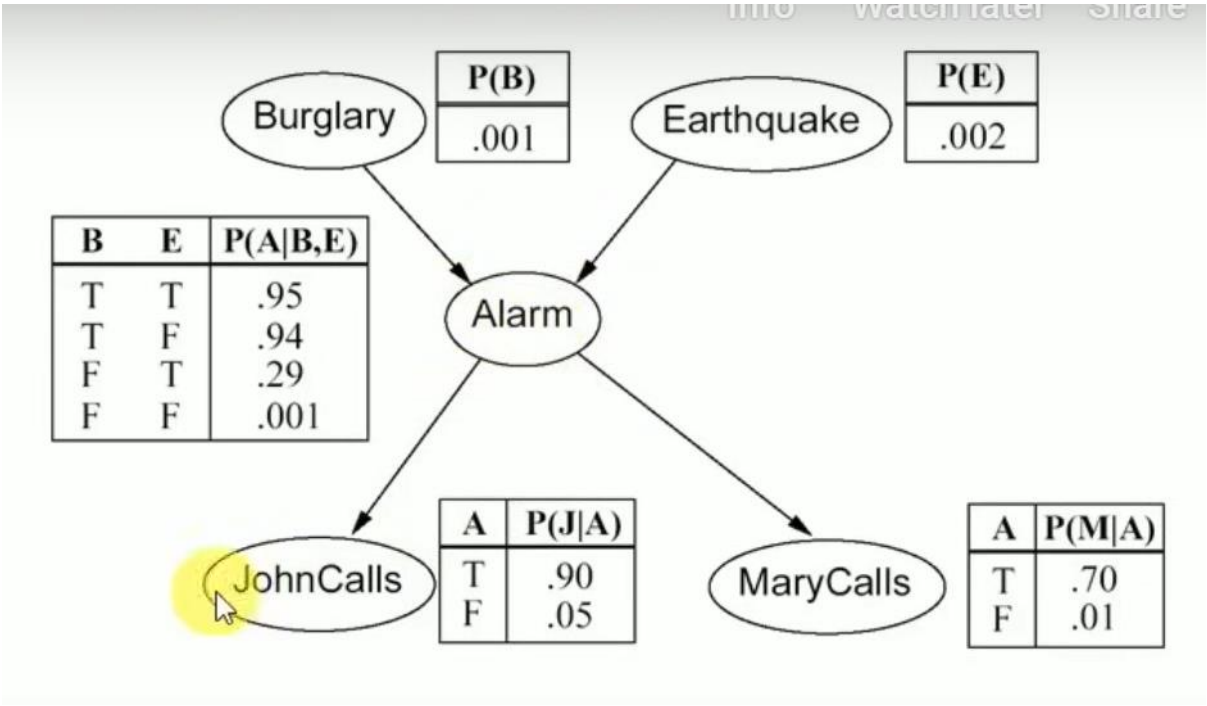
$$P(j \wedge m \wedge a \wedge \neg b \wedge \neg e) = P(j \mid a)\, P(m \mid a)\, P(a \mid \neg b, \neg e)\, P(\neg b)\, P(\neg e)$$

$$= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998$$

$$= 0.00062$$

2. What is the probability that John call?

Consider the given network



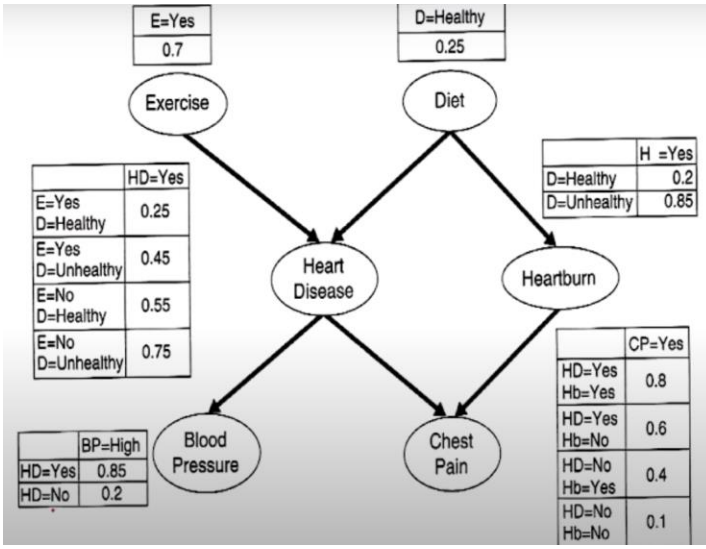
Solution:

$P(j) = P(j \mid a)\,P(a) + P(j \mid \neg a)\,P(\neg a)$

$= P(j|a)\{P(a|b,e)*P(b,e)+P(a|\neg b,e)*P(\neg b,e)+P(a|b,\neg e)*P(b,\neg e)+P(a|\neg b,\neg e)*P(\neg b,\neg e)\}$

$+ P(j|\neg a)\{P(\neg a|b,e)*P(b,e)+P(\neg a|\neg b,e)*P(\neg b,e)+P(\neg a|b,\neg e)*P(b,\neg e)+P(\neg a|\neg b,\neg e)*P(\neg b,\neg e)\}$

$= 0.90 * 0.00252 + 0.05 * 0.9974 = 0.0521$



- Suppose we get to know that the new patient has **high blood pressure.**
- What's the probability he has **heart disease** under this condition?

$$P(hd \mid bp) = \alpha \sum_e \sum_d \sum_h \sum_{cp} P(hd, e, d, h, cp, bp)$$

$$= \alpha \sum_e \sum_d \sum_h \sum_{cp} P(hd \mid e,d) P(h \mid d) P(cp \mid hd, h) P(bp \mid hd) P(e) P(d)$$

$$= \alpha P(bp \mid hd) \sum_e P(e) \sum_d P(hd \mid e,d) P(d) \sum_h P(h \mid d) \sum_{cp} P(cp \mid hd, h)$$

$$= \alpha P(bp \mid hd) \sum_e P(e) \sum_d P(hd \mid e,d) P(d)$$

$$= \alpha P(bp \mid hd) P(e) \left(P(hd \mid e,d) P(d) + P(hd \mid e, \neg d) P(\neg d)\right)$$
$$+ \alpha P(bp \mid hd) P(\neg e) \left(P(hd \mid \neg e, d) P(d) + P(hd \mid \neg e, \neg d) P(\neg d)\right)$$

$$= \alpha * 0.85 * 0.7 * (0.25 * 0.25 + 0.45 * 0.75) + \alpha * 0.85 * 0.3 * (0.55 * 0.25 + 0.75 * 0.75)$$

$$= \alpha * 0.4165$$

$$P(\neg hd \mid bp) = \alpha \sum_e \sum_d \sum_h \sum_{cp} P(\neg hd, e, d, h, cp, bp)$$

$$= \alpha \sum_e \sum_d \sum_h \sum_{cp} P(\neg hd \mid e,d) P(h \mid d) P(cp \mid \neg hd, h) P(bp \mid \neg hd) P(e) P(d)$$

$$= \alpha P(bp \mid \neg hd) \sum_e P(e) \sum_d P(\neg hd \mid e,d) P(d) \sum_h P(h \mid d) \sum_{cp} P(cp \mid \neg hd, h)$$

$$= \alpha P(bp \mid \neg hd) \sum_e P(e) \sum_d P(\neg hd \mid e,d) P(d)$$

$$= \alpha P(bp \mid \neg hd) P(e) \left(P(\neg hd \mid e,d) P(d) + P(\neg hd \mid e, \neg d) P(\neg d)\right)$$
$$+ \alpha P(bp \mid \neg hd) P(\neg e) \left(P(\neg hd \mid \neg e, d) P(d) + P(\neg hd \mid \neg e, \neg d) P(\neg d)\right)$$

$$= \alpha * 0.2 * 0.7 * (0.75 * 0.25 + 0.55 * 0.75) + \alpha * 0.2 * 0.3 * (0.45 * 0.25 + 0.25 * 0.75)$$

$$= \alpha * 0.102$$

$$P(hd \mid bp) = \alpha * 0.4165 \qquad P(\neg hd \mid bp) = \alpha * 0.102$$

$$\alpha = \frac{1}{0.4165 + 0.1020} = \frac{1}{0.5185}$$

$$P(hd \mid bp) = \alpha * 0.4165 = 0.8033$$

$$P(\neg hd \mid bp) = \alpha * 0.102 = 0.1967$$

Support vector machines (SVM)

Support vector machines are a set of supervised learning methods used for

- classification,
- regression,
- and outliers detection.

All of these are common tasks in machine learning.

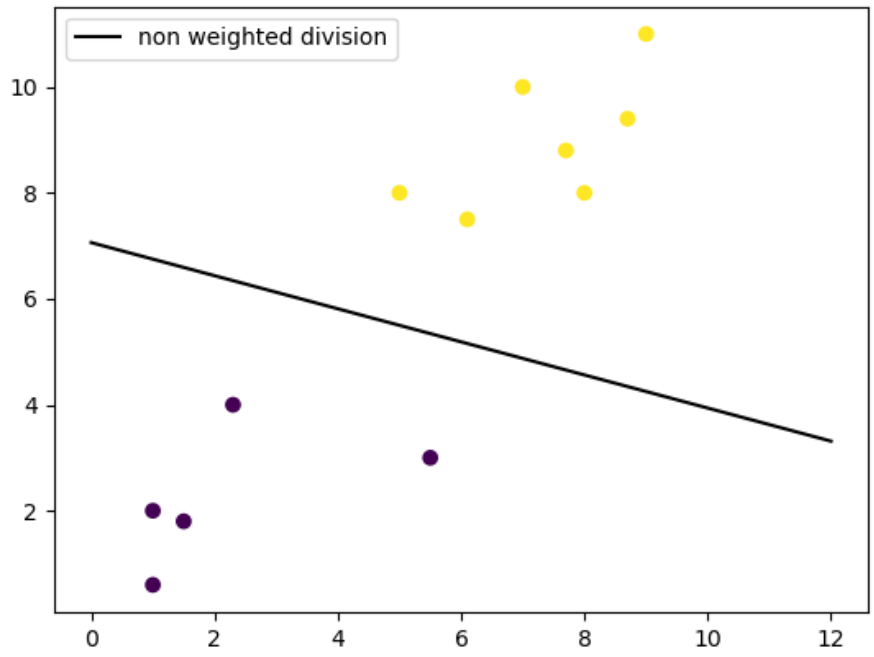
You can use them to detect cancerous cells based on millions of images or you can use them to predict future driving routes with a well-fitted regression model.

There are specific types of SVMs you can use for particular machine learning problems, like support vector regression (SVR) which is an extension of support vector classification (SVC).

A simple linear SVM classifier works by making a straight line between two classes. That means all of the data points on one side of the line will represent a category and the data points on the other side of the line will be put into a different category. This means there can be an infinite number of lines to choose from.

Basically you have some data points on a grid. You're trying to separate these data points by the category they should fit in, but you don't want to have any data in the wrong category. That means you're trying to find the line between the two closest points that keeps the other data points separated.

So the two closest data points give you the support vectors you'll use to find that line. That line is called the decision boundary.



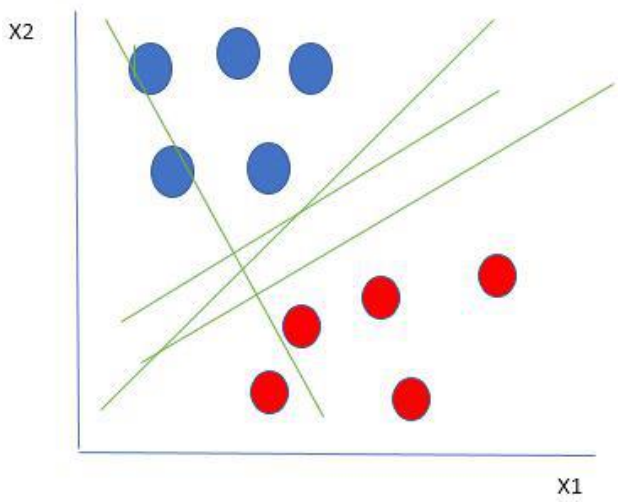


Fig:Linearly Separable Data points

From the figure above it's very clear that there are multiple lines (our hyperplane here is a line because we are considering only two input features x1, x2) that segregate our data points or do a classification between red and blue circles.

One reasonable choice for the **best hyperplane** in a **Support Vector Machine (SVM)** is the one that maximizes the **separation margin** between the two classes. The **maximum-margin hyperplane**, also referred to as the **hard margin**, is selected based on maximizing the distance between the hyperplane and the nearest data point on each side.

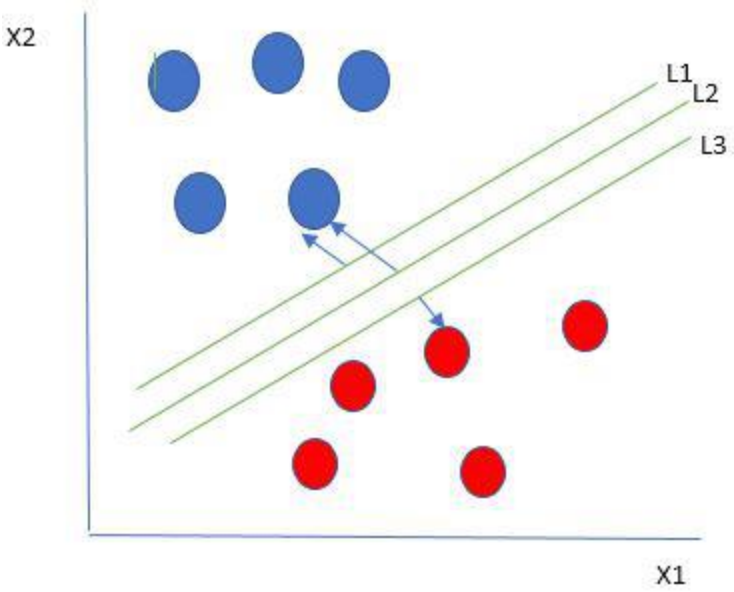
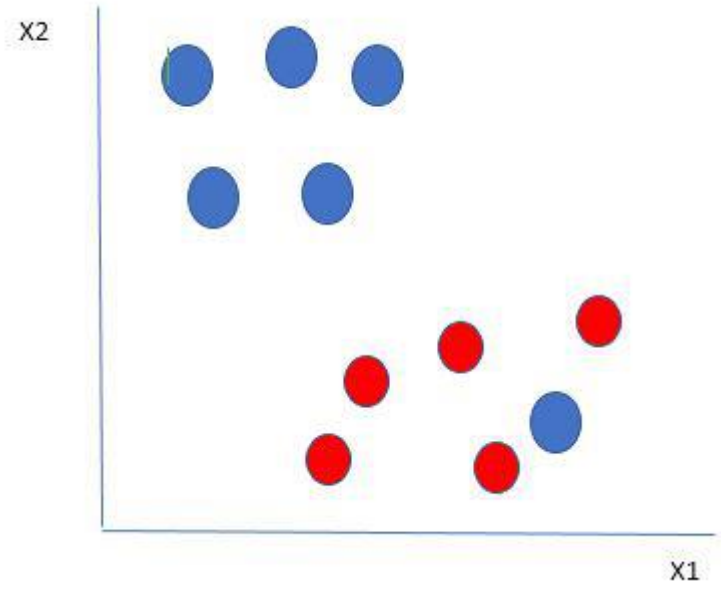
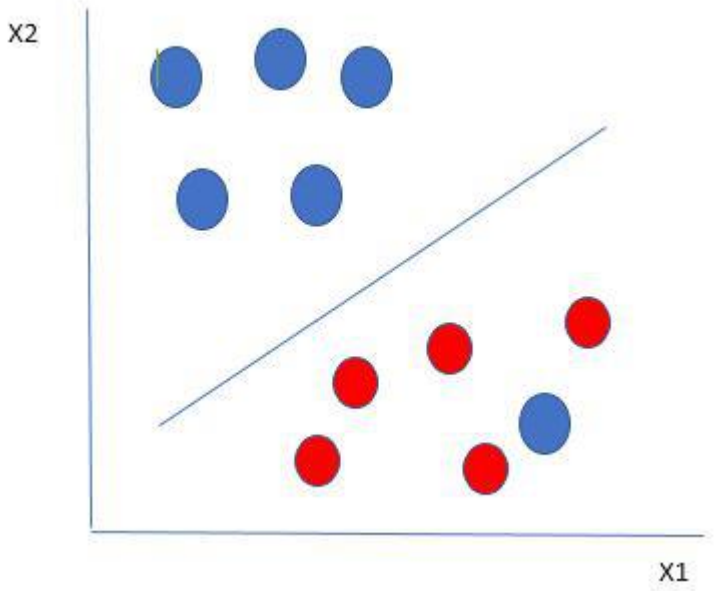


Fig:Multiple hyperplanes separate the data from two classes

So we choose the hyperplane whose distance from it to the nearest data point on each side is maximized. If such a hyperplane exists it is known as the **maximum-margin hyperplane/hard margin**. So from the above figure, we choose L2.



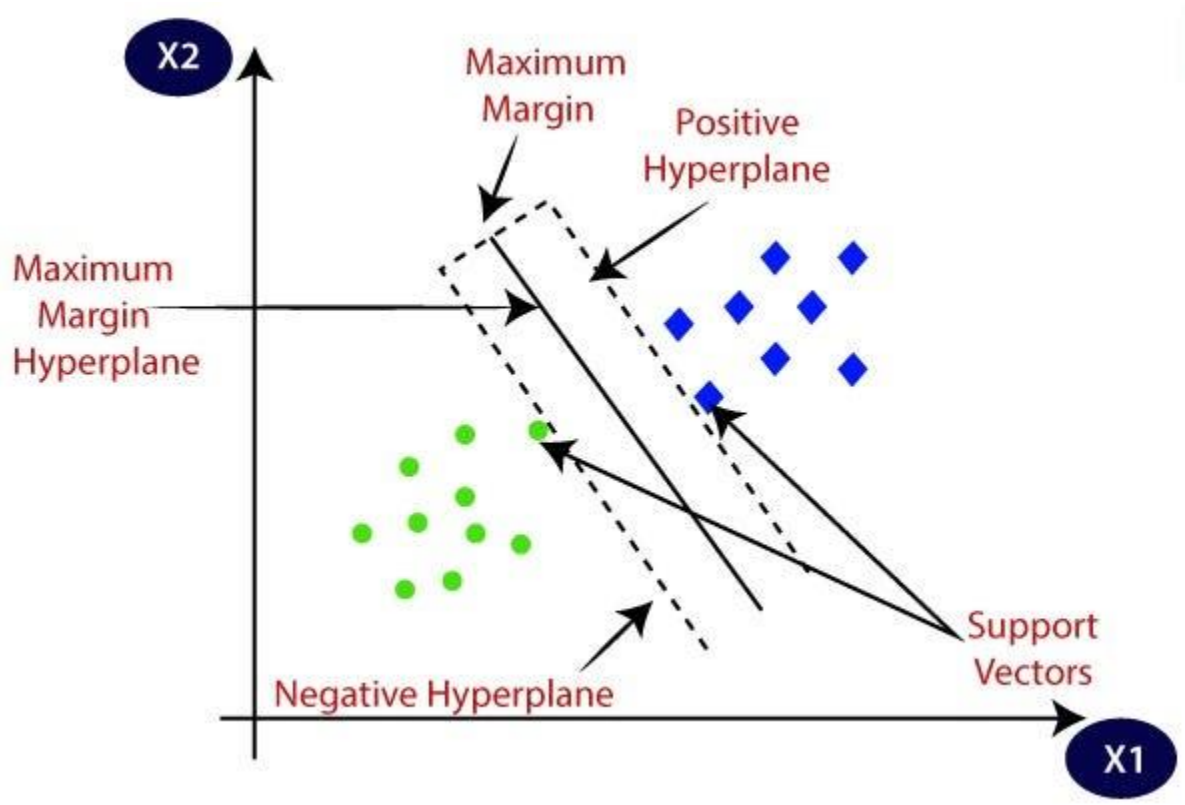
Selecting hyperplane for data with outlier Here we have one blue ball in the boundary of the red ball. So how does SVM classify the data? It's simple! The blue ball in the boundary of red ones is an outlier of blue balls. The SVM algorithm has the characteristics to ignore the outlier and finds the best hyperplane that maximizes the margin. SVM is robust to outliers.



Hyperplane which is the most optimized one

Support Vector Machine Terminology

- **Hyperplane**: The **hyperplane** is the decision boundary used to separate data points of different classes in a feature space. For **linear classification**, this is a linear equation represented as wx+b=0.
- **Support Vectors**: **Support vectors** are the closest data points to the hyperplane. These points are critical in determining the hyperplane and the margin in **Support Vector Machine (SVM)**.
- **Margin**: The **margin** refers to the distance between the **support vector** and the hyperplane. The primary goal of the SVM algorithm is to maximize this margin, as a wider margin typically results in better classification performance.
- **Kernel**: The **kernel** is a mathematical function used in SVM to map input data into a higher-dimensional feature space. This allows the SVM to find a hyperplane in cases where data points are not linearly separable in the original space. Common **kernel functions** include linear, polynomial, radial basis function (RBF), and sigmoid.



Types of SVMs

There are two different types of SVMs,

- **Linear SVM**: When the data is perfectly linearly separable only then we can use Linear SVM. Perfectly linearly separable means that the data points can be classified into 2 classes by using a single straight line(if 2D).
- **Non-Linear SVM**: When the data is not linearly separable then we can use Non-Linear SVM, which means when the data points cannot be separated into 2 classes by using a straight line (if 2D) then we use some advanced techniques like kernel tricks to classify them. In most real-world applications we do not find linearly separable datapoints hence we use kernel trick to solve them.

Example:

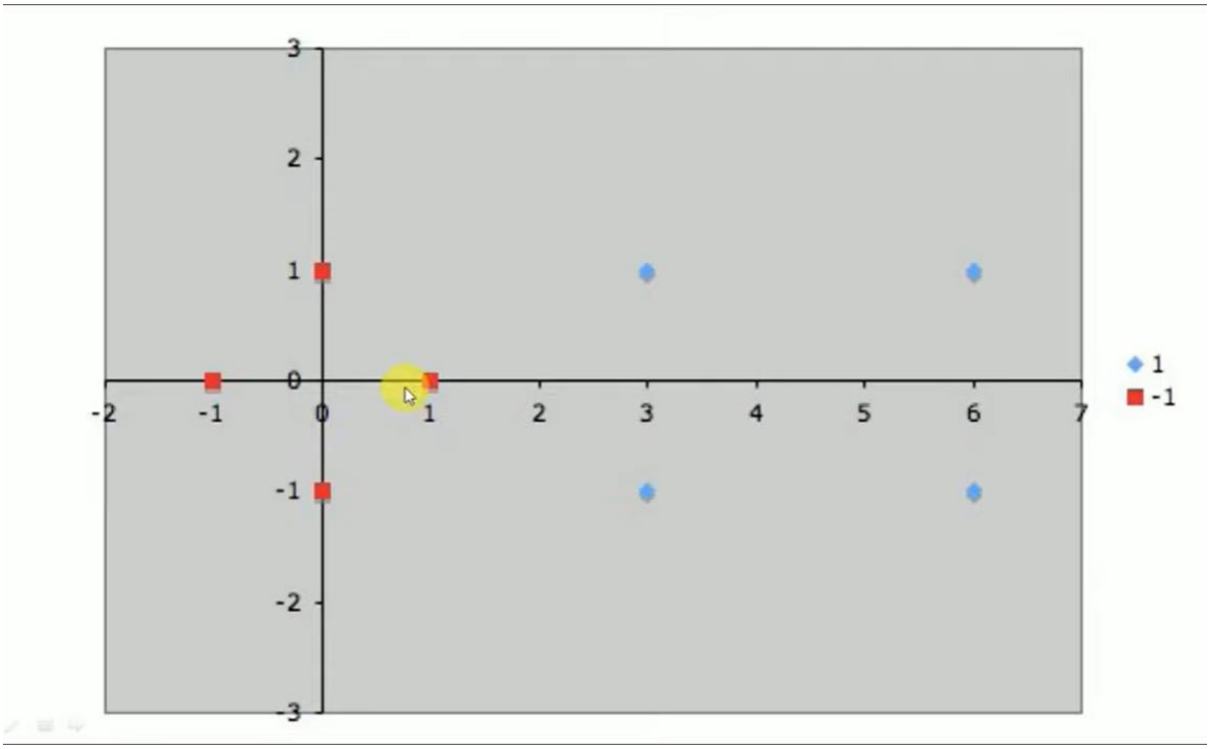
Support Vector Machine - Linear Example Solved

Suppose we are given the following positively labeled data points,

$$\left\{\begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix}\right\}$$

and the following negatively labeled data points,

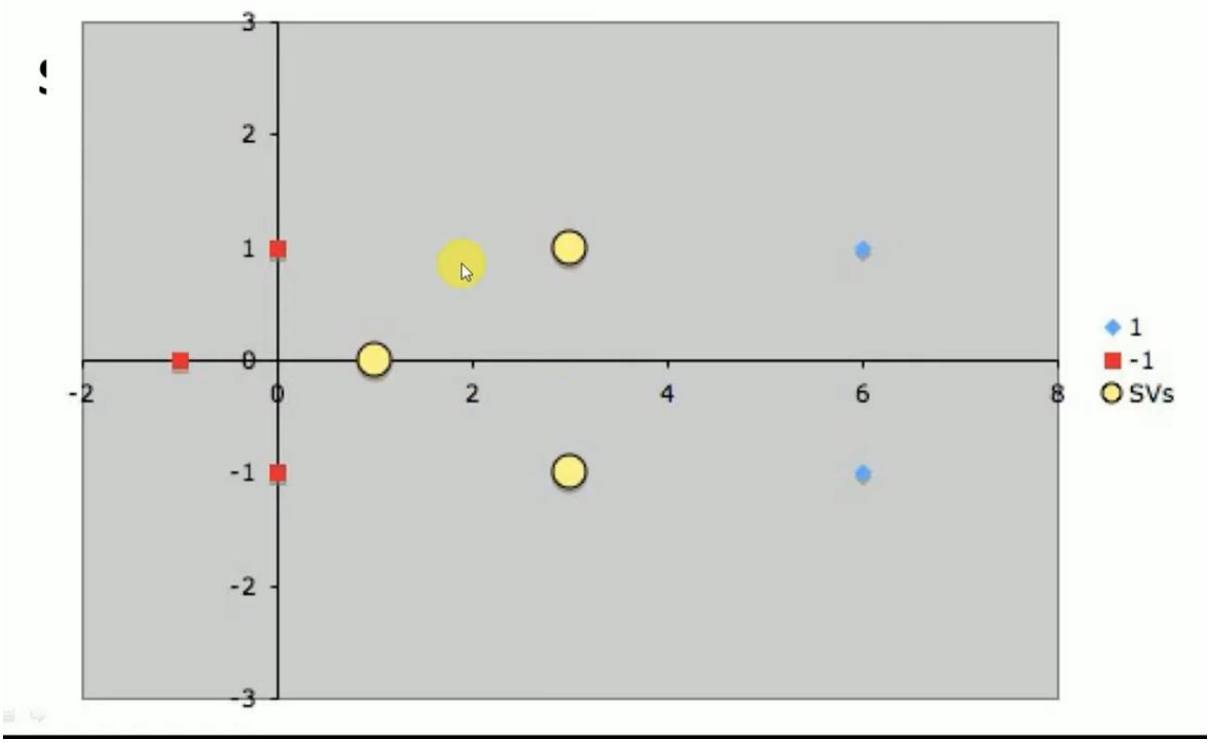
$$\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix}\right\}$$



Support Vector Machine - Linear Example Solved

- By inspection, it should be obvious that there are **three** support vectors,

$$\left\{ s_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, s_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, s_3 = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \right\}$$



Support Vector Machine - Linear Example Solved

- Each vector is augmented with a 1 as a bias input
- So, $s_1 = \begin{pmatrix}1\\0\end{pmatrix}$, then $\tilde{s_1} = \begin{pmatrix}1\\0\\1\end{pmatrix}$
- Similarly,
- $s_2 = \begin{pmatrix}3\\1\end{pmatrix}$, then $\tilde{s_2} = \begin{pmatrix}3\\1\\1\end{pmatrix}$ *and* $s_3 = \begin{pmatrix}3\\-1\end{pmatrix}$, then $\tilde{s_3} = \begin{pmatrix}3\\-1\\1\end{pmatrix}$

We need to calculate weight vector. To calculate this we have to find alph1 and alha2 and alpha 3 values

Support Vector Machine - Linear Example Solved

$$\alpha_1 \tilde{s_1} \cdot \tilde{s_1} + \alpha_2 \tilde{s_2} \cdot \tilde{s_1} + \alpha_3 \tilde{s_3} \cdot \tilde{s_1} = -1$$

In this equation S1 is common so equate tp -1

$$\alpha_1 \tilde{s_1} \cdot \tilde{s_2} + \alpha_2 \tilde{s_2} \cdot \tilde{s_2} + \alpha_3 \tilde{s_3} \cdot \tilde{s_2} = +1$$

In this equation S2 is common so equate tp 1

$$\alpha_1 \tilde{s_1} \cdot \tilde{s_3} + \alpha_2 \tilde{s_2} \cdot \tilde{s_3} + \alpha_3 \tilde{s_3} \cdot \tilde{s_3} = +1$$

In this equation S3 is common so equate tp 1

$$\begin{aligned}
\alpha_1 \tilde{s_1}\cdot\tilde{s_1} + \alpha_2 \tilde{s_2}\cdot\tilde{s_1} + \alpha_3 \tilde{s_3}\cdot\tilde{s_1} &= -1 \\
\alpha_1 \tilde{s_1}\cdot\tilde{s_2} + \alpha_2 \tilde{s_2}\cdot\tilde{s_2} + \alpha_3 \tilde{s_3}\cdot\tilde{s_2} &= +1 \\
\alpha_1 \tilde{s_1}\cdot\tilde{s_3} + \alpha_2 \tilde{s_2}\cdot\tilde{s_3} + \alpha_3 \tilde{s_3}\cdot\tilde{s_3} &= +1
\end{aligned}$$

$$\alpha_1(1+0+1)+\alpha_2(3+0+1)+\alpha_3(3+0+1) = -1$$

$$\alpha_1(3+0+1)+\alpha_2(9+1+1)+\alpha_3(9-1+1) = 1$$

$$\alpha_1(3+0+1)+\alpha_2(9-1+1)+\alpha_3(9+1+1) = 1$$

$$\alpha_1\begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix}+\alpha_2\begin{pmatrix}3\\1\\1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix}+\alpha_3\begin{pmatrix}3\\-1\\1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix} = -1$$

$$\alpha_1\begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}3\\1\\1\end{pmatrix}+\alpha_2\begin{pmatrix}3\\1\\1\end{pmatrix}\begin{pmatrix}3\\1\\1\end{pmatrix}+\alpha_3\begin{pmatrix}3\\-1\\1\end{pmatrix}\begin{pmatrix}3\\1\\1\end{pmatrix} = 1$$

$$\alpha_1\begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}3\\-1\\1\end{pmatrix}+\alpha_2\begin{pmatrix}3\\1\\1\end{pmatrix}\begin{pmatrix}3\\-1\\1\end{pmatrix}+\alpha_3\begin{pmatrix}3\\-1\\1\end{pmatrix}\begin{pmatrix}3\\-1\\1\end{pmatrix} = 1$$

$$2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1$$

$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1$$

$$4\alpha_1 + 9\alpha_2 + 11\alpha_3 = 1$$

$2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1 \quad -(1)$

$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1 \quad -(2)$

$4\alpha_1 + 9\alpha_2 + 11\alpha_3 = 1 \quad -(3)$

$(2) - (3)$

$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1$

$-4\alpha_1 \pm 9\alpha_2 \pm 11\alpha_3 = 1$

$2\alpha_2 - 2\alpha_3 = 0$

$2\alpha_2 = 2\alpha_3$

$\alpha_2 = \alpha_3 \quad -(4)$

Substitute (4) in (1)

$2\alpha_1 + 4\alpha_2 + 4\alpha_2 = -1$

~~$4\alpha_1 + 11\alpha_2 + 9\alpha_2 = 1$~~

$2\alpha_1 + 8\alpha_2 = -1$

$2\alpha_1 = -1 - 8\alpha_2$

$\alpha_1 = \frac{-1 - 8\alpha_2}{2} \quad -(5)$

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Substitute (5) in (2)

$$2\left(\frac{-1-8\alpha_2}{2}\right)+11\alpha_2+9\alpha_2=1$$

$$-2-16\alpha_2+11\alpha_2+9\alpha_2=1$$

$$4\alpha_2=3$$

$$\boxed{\alpha_2=3/4=0.75} \quad -(6)$$

Substitute (6) in (5)

$$\alpha_1=\frac{-1-8\times\frac{3}{4}}{2}$$

$$=\frac{-1-6}{2}$$

$$=-\frac{7}{2}$$

$$\therefore \boxed{\alpha_1=-3.5}$$

$$\boxed{\alpha_3=\alpha_2=0.75}$$

WEIGHT VECTOR

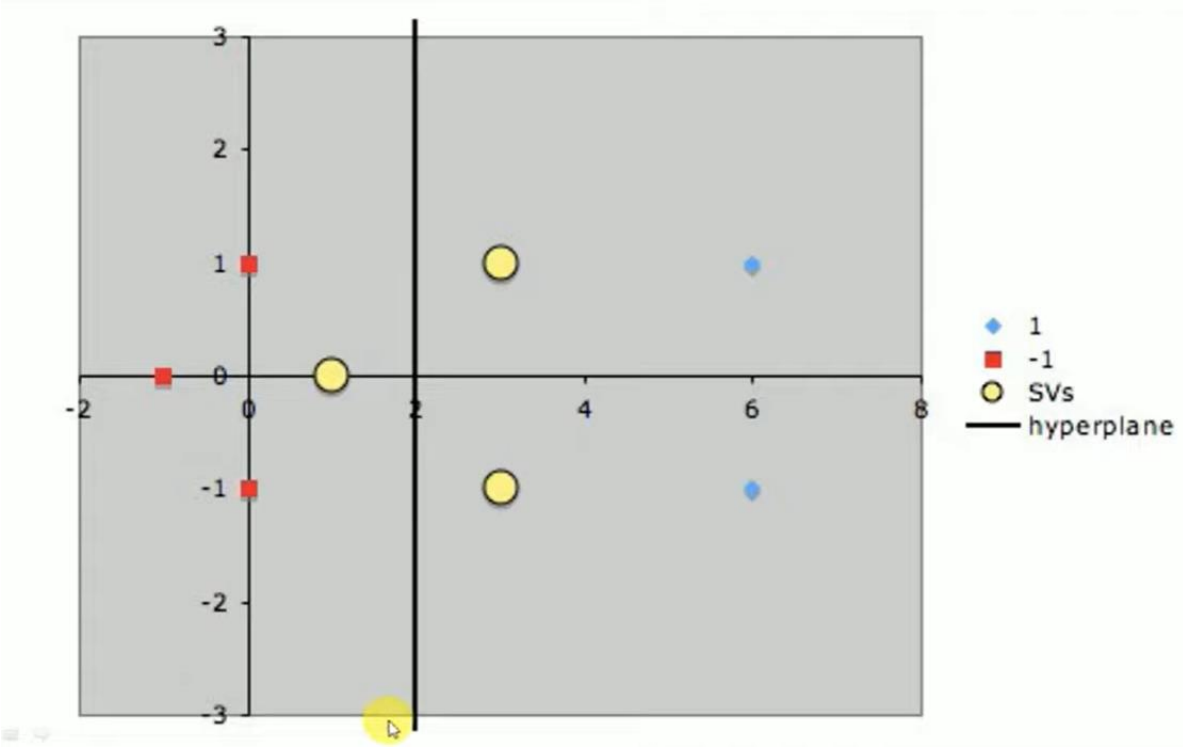
i is the nearest data points. in our example we have 3 nearest data points

$$\tilde{w} = \sum_i \alpha_i \tilde{s}_i$$

$$= -3.5 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 0.75 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + 0.75 \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

- Finally, remembering that our vectors are augmented with a bias.
- We can equate the last entry in $\tilde{w}$ as the hyperplane offset b and write the separating
- Hyperplane equation $\boldsymbol{y = wx + b}$
- with $\boldsymbol{w} = \binom{1}{0}$ and $\boldsymbol{b = -2}$.



Now, for a new data point (2, -2). Classify it

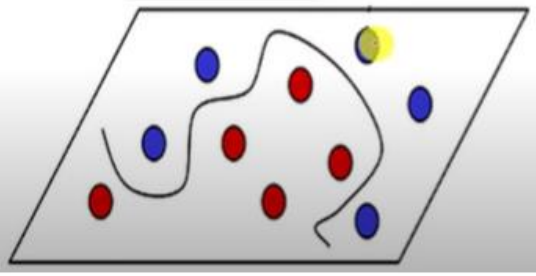
$wx+b \geq 0$ will belong to class +1

$wx+b < 0$ will belong to class -1

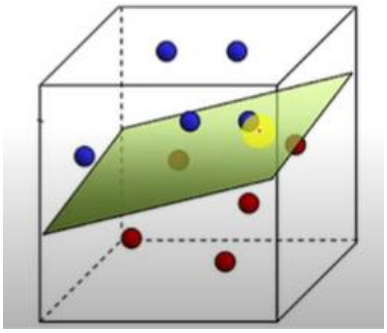
$w=\begin{pmatrix}1\\0\end{pmatrix}\begin{pmatrix}2\\-2\end{pmatrix}-2 \Rightarrow 2-0-2=0$ i.e. class +1

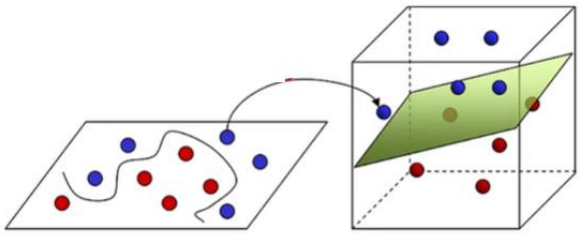
SVM Kernels:

- In machine learning applications, the data can be text, image, or video.
- So, there is a need to extract features from these data prior to classification.
- Hence, in the real world, many classification models are complex and mostly require non-linear hyperplanes.



Convert 2D data to 3D data. Now we can separate two classes.





Example to convert 2D to 3D

- For example, one mapping function $\emptyset: \boldsymbol{R^2} \rightarrow \boldsymbol{R^3}$ used to transform a 2D data to 3D data is given as follows:

$$\emptyset(\boldsymbol{x}, \boldsymbol{y}) = (\boldsymbol{x^2}, \sqrt{\boldsymbol{2}}\boldsymbol{xy}, \boldsymbol{y^2})$$

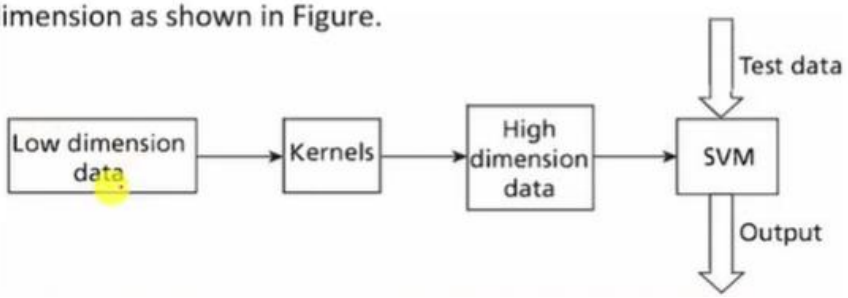
- Consider a point (2, 3) in 2D space, if you apply above mapping function we can convert it into 3D space and it looks like this,
- Here $x = 2\ and\ y = 3$,
- Hence point in 3D space is:

$$\left(\boldsymbol{2^2}, \sqrt{\boldsymbol{2}} * \boldsymbol{2} * \boldsymbol{3}, \boldsymbol{3^2}\right) = (\boldsymbol{4}, \boldsymbol{6\sqrt{2}}, \boldsymbol{9})$$

- While mapping functions play an important role, there are many disadvantages, as mapping involves more computations and learning costs.
- Also, the disadvantages of transformations are that there is no generalized thumb rule available describing what transformations should be applied and if the data is large, mapping process takes huge amount of time.

- In real applications, there might be many features in the data and applying transformations that involve many polynomial combinations of these features will lead to extremely high and impractical computational costs.
- In this context, only kernels are useful.
- Kernels are used to compute the value without transforming the data.

- **What is a Kernel?**
- Kernels are a set of functions used to transform data from lower dimension to higher dimension and to manipulate data using dot product at higher dimensions.
- The use of kernels is to apply transformation to data and perform classification at the higher dimension as shown in Figure.



- **Kernel Trick for 2nd degree Polynomial Mapping** $\emptyset(x, y) = (x^2, \sqrt{2}xy, y^2)$

$$\phi(\mathbf{a})^T \cdot \phi(\mathbf{b}) = \begin{pmatrix} a_1^2 \\ \sqrt{2}\,a_1 a_2 \\ a_2^2 \end{pmatrix}^T \cdot \begin{pmatrix} b_1^2 \\ \sqrt{2}\,b_1 b_2 \\ b_2^2 \end{pmatrix} = a_1^2 b_1^2 + 2a_1 b_1 a_2 b_2 + a_2^2 b_2^2$$

$$= (a_1 b_1 + a_2 b_2)^2 = \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}^T \cdot \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}\right)^2 = (\mathbf{a}^T \cdot \mathbf{b})^2$$

Types of Kernels

- Linear Kernel
- Polynomial Kernel
- Exponential Kernel
- Homogeneous Kernel
- Inhomogeneous Kernel
- Gaussian Kernel
- Sigmoid Kernel
- Radial-basis function kernel

Linear Kernel

- Linear kernels are of the type

$$k(x, y) = x^T . y$$

- where x and y are two vectors.
- Therefore $k(x, y) = \emptyset(x).\emptyset(y) = x^T . y$

Polynomial Kernel

- Polynomial kernels are of the type:

$$k(x, y) = (x^T y)^q$$

- This is called homogeneous kernel.
- Here, q is the degree of the polynomial.
- If q = 2 then it is called **quadratic kernel**.

Polynomial Kernel

- For inhomogeneous kernels, this is given as:

$$k(x, y) = (c + x^T y)^q$$

- Here c is a constant and q is the degree of the polynomial.
- If c is zero and degree is one, the polynomial kernel is reduced to a linear kernel.
- The value of degree q should be optimal as more degree may lead to overfitting.

Example: Linear, Homogeneous and Inhomogeneous

- Consider two data points $x = \begin{pmatrix}1\\2\end{pmatrix}$ and $y = (2,3)$ with c =1. Apply linear, homogeneous and inhomogeneous kernels.
- **Solution:**
- The kernel is given by $k(x,y) = (x^T y)^q$
- If q = 1 the it is called linear kernel
- $k(x,y) = \left(\begin{pmatrix}1\\2\end{pmatrix}^T (2,3)\right)^1 = 8$

(1, 2).(2,3)=(1*2+2*3)=8

Homogeneous Kernel or quadratic Kernel

- The kernel is given by $k(x,y) = (x^T y)^q$
- If q = 2 the it is called homogeneous or quadratic kernel
- $k(x,y) = \left(\begin{pmatrix}1\\2\end{pmatrix}^T (2,3)\right)^2 = 8^2 = 64$

Inomogeneous Kernel

- The kernel is given by $k(x,y) = (x^T y)^q$
- If q = 2 and c =1 the it is called inhomogeneous kernel
- $k(x,y) = (c + x^T y)^q = \left(1 + \begin{pmatrix}1\\2\end{pmatrix}^T (2,3)\right)^2 = (1+8)^2 = 81$

Gaussian Kernel

- Radial Basis Functions (RBFs) or Gaussian kernels are extremely useful in SVM.
- The RBF function is shown as below:

$$k(x,y) = e^{\frac{-(x-y)^2}{2\sigma^2}}$$

- Here, y is an important parameter. If y is small, then the RBF is similar to linear SVM and if y is large, then the kernel is influenced by more support vectors.

Example:

- Consider two data points x = (1. 2) and y = (2, 3) with $\sigma = 1$.
- Apply RBF kernel and find the value of RBF kernel for these points.

$$k(x, y) = e^{\frac{-(x-y)^2}{2\sigma^2}}$$

- Substitute the value of x and y in RBF kernel.
- The squared distance between the points (1, 2) and (2, 3) is given as:

$$(1 - 2)^2 + (2 - 3)^2 = 2$$

- If $\sigma = 1$, then $k(x, y) = e^{\left\{-\frac{2}{2}\right\}} = e^{-1} = 0.3679$.

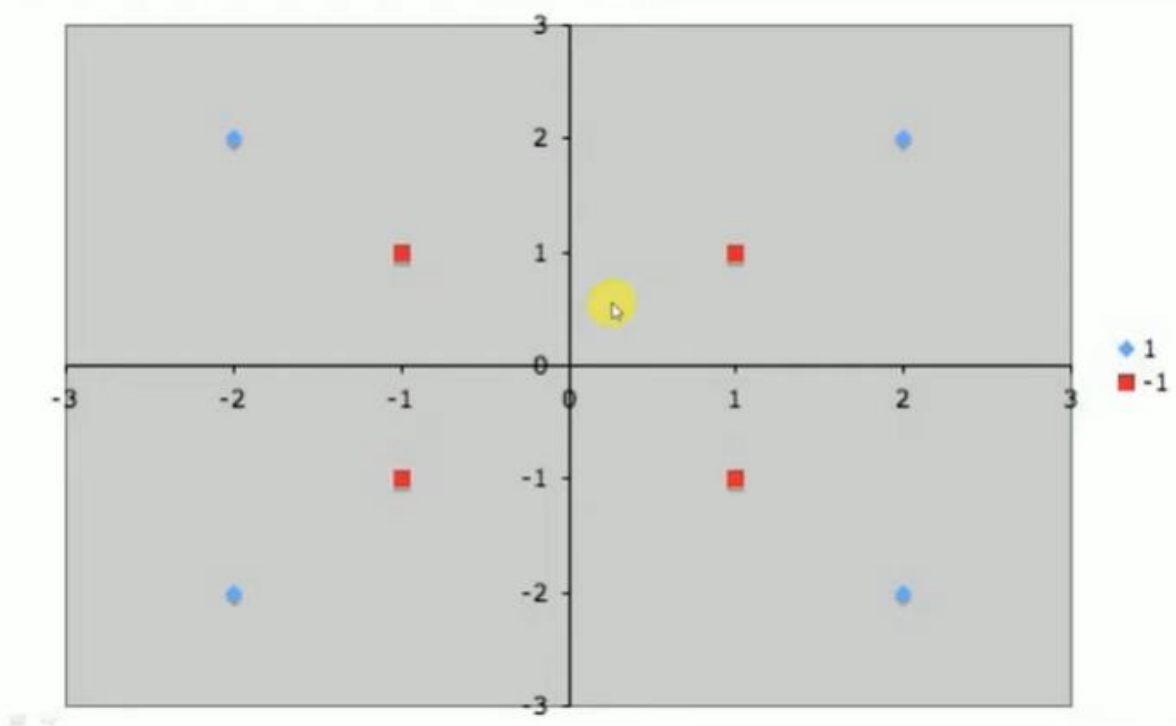
Support Vector Machine – Non Linear Example Solved

Suppose we are given the following positively labeled data points,

$$\left\{\begin{pmatrix} 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \end{pmatrix}\right\}$$

and the following negatively labeled data points,

$$\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\}$$



- Our goal, again, is to discover a separating hyperplane that accurately discriminates the two classes.
- Of course, it is obvious that no such hyperplane exists in the input space
- Therefore, we must use a nonlinear SVM (that is, we need to convert data from one feature space to another.

$$\Phi_1 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{cases} \begin{pmatrix} 4 - x_2 + |x_1 - x_2| \\ 4 - x_1 + |x_1 - x_2| \end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} > 2 \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} & \text{otherwise} \end{cases}$$

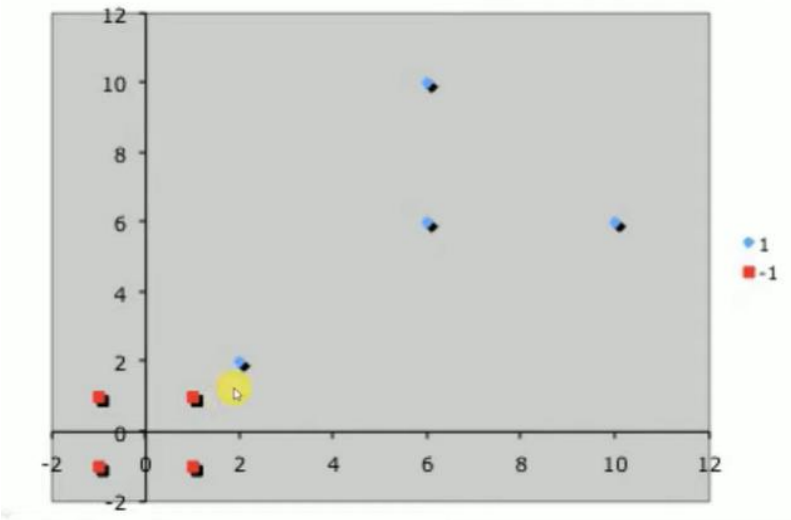
$$\Phi_1 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{cases} \begin{pmatrix} 4 - x_2 + |x_1 - x_2| \\ 4 - x_1 + |x_1 - x_2| \end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} > 2 \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} & \text{otherwise} \end{cases}$$

Positive Examples

$$\left\{ \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \end{pmatrix} \right\} \rightarrow \left\{ \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 10 \\ 6 \end{pmatrix}, \begin{pmatrix} 6 \\ 6 \end{pmatrix}, \begin{pmatrix} 6 \\ 10 \end{pmatrix} \right\}$$

Negative Examples

$$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\} \rightarrow \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$$



- Now we can easily identify the support vectors,

$$\left\{ s_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, s_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}$$

- Each vector is augmented with a 1 as a bias input

$$\tilde{s_1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \qquad \tilde{s_2} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

$$\alpha_1 \tilde{s_1} \cdot \tilde{s_1} + \alpha_2 \tilde{s_2} \cdot \tilde{s_1} = -1 \qquad \alpha_1(1+1+1)+\alpha_2(2+2+1) = -1$$
$$\alpha_1 \tilde{s_1} \cdot \tilde{s_2} + \alpha_2 \tilde{s_2} \cdot \tilde{s_2} = +1 \qquad \alpha_1(2+2+1)+\alpha_2(4+4+1) = 1$$

$$\alpha_1 \begin{pmatrix}1\\1\\1\end{pmatrix}\begin{pmatrix}1\\1\\1\end{pmatrix} + \alpha_2 \begin{pmatrix}2\\2\\1\end{pmatrix}\begin{pmatrix}1\\1\\1\end{pmatrix} = -1 \qquad 3\alpha_1 + 5\alpha_2 = -1$$

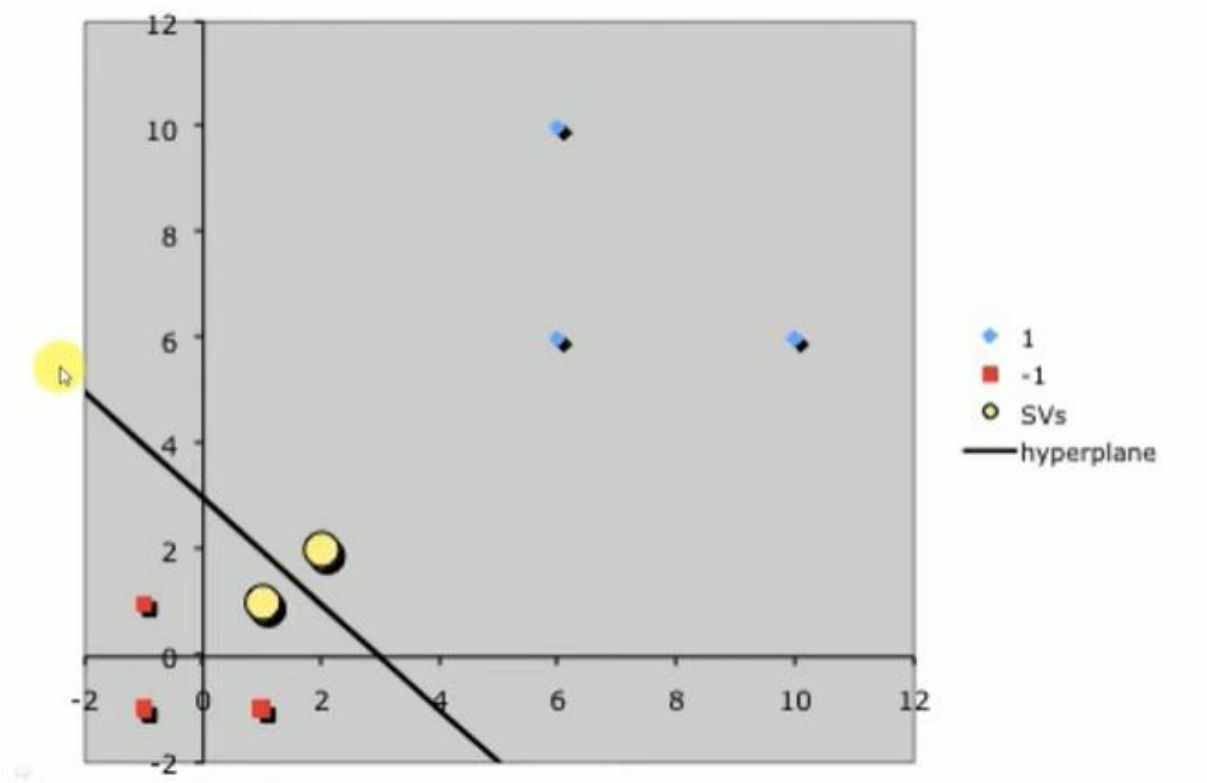
$$5\alpha_1 + 9\alpha_2 = 1$$

$$\alpha_1 \begin{pmatrix}1\\1\\1\end{pmatrix}\begin{pmatrix}2\\2\\1\end{pmatrix} + \alpha_2 \begin{pmatrix}2\\2\\1\end{pmatrix}\begin{pmatrix}2\\2\\1\end{pmatrix} = 1$$

$$\alpha_1 = -7$$
$$\alpha_2 = 4$$

$$\tilde{w} = \sum_i \alpha_i \tilde{s_i} = -7\begin{pmatrix}1\\1\\1\end{pmatrix} + 4\begin{pmatrix}2\\2\\1\end{pmatrix}$$
$$= \begin{pmatrix}1\\1\\-3\end{pmatrix}$$

- Finally, remembering that our vectors are augmented with a bias.
- We can equate the last entry in $\tilde{w}$ as the hyperplane offset b and write the separating
- Hyperplane equation $\boldsymbol{y = wx + b}$
- with $\boldsymbol{w} = \binom{1}{1}$ and $\boldsymbol{b = -3}$.


12
10
8
6
4
2
0
-2
-2
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hyperplane